Carter Cripe

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EDUCATION

Oregon State University · September 2023 - June 2027

B.S. Computer Science, 3.72 GPA · *Dean's List x8* · Corvallis, OR

- **Relevant Coursework:** Machine Learning; Data Mining; Artificial Intelligence; Data Science; Mathematical Statistics; Linear Algebra; Data Structures and Algorithms
- **Extracurriculars:** Captain and officer, OSU Ultimate Frisbee team (#7 nationally ranked); Hackathon and Cybersecurity clubs

EXPERIENCE

Thermo Fisher Scientific · April 2026 - September 2026

Software Engineering Intern · Data, AI, Visualization (DAIV) team · Hillsboro, OR

- Assisted development of a new AI model for analyzing semiconductor microscope images and benchmarked it against the company's existing approach: 37% more accurate, nearly doubling success rate (43% to 78%)
- Showed team's model needed about 68% less labeled training data, achieving successful quality in 15 of 17 image classes with three or fewer labeled examples (existing approach: 4 of 17)
- Designed and built a modular Python framework for repeatable segmentation experiments, now used as team's reusable benchmarking infrastructure, plus an agent skill teaching AI coding agents its conventions
- Expanded DAIV AI model training and inference servers: a shared code library for both, training settings adjustable through an API, and end-to-end integration with microscope's image-analysis software

PROJECTS

Persistent 3D Scene Understanding for Robotics · PNNL Capstone · Oct. 2026 - Present

- Building a persistent 3D world model service for a *Pacific Northwest National Laboratory* sponsored senior capstone, fusing RGB-D camera input with SAM 2–based segmentation to track laboratory objects over time
- Implementing change detection and occlusion recovery to keep robot's world state accurate when objects move or hide, published through a reusable ROS2 interface

Cicadas - Contributor · github.com/ecodan/cicadas · November 2025 - Present

- Open source framework for keeping specs aligned with code as AI agents build and regenerates documentation from what shipped; Used on my own projects and presented it to a Thermo Fisher R&D team
- Architected and merged a scope gate letting AI agents right-size requests from full initiatives to lightweight tweaks
- Framework used by 2 Fortune 500 corporations to manage codebase documentation and ensure safe AI coding

Field View - Ultimate Frisbee spacing model · ultipedia.cartercripe.com/fieldview/explore · July 2026 - Present

- Developed an interactive model shading each location on the field by whether it is 'open' or 'defended', computed live from disc flight time and player kinematics as players are dragged
- Created Ultipedia (React, TypeScript, FastAPI, Supabase), a mobile-first web app around it, collecting coaches' drills and strategies for an ultimate-frisbee encyclopedia

UltiLines - Mobile app for Ultimate coaches · October 2025

- Constructed a cross-platform app (React Native, TypeScript) for building lines, recording live game stats, and generating post-game summaries; released on the App Store with 50+ downloads and 15+ teams

SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, React, SQL, HTML/CSS

Machine Learning & Computer Vision: PyTorch, computer vision, image segmentation (SAM, CNNs), few-shot learning, transfer learning, model evaluation, data augmentation, vision-language models, numpy, pandas

AI & LLM Tools: Claude Code, Codex, Cursor, QA testing, agentic workflows, spec-driven development

Backend & Tools: Flask, FastAPI, Next.js, REST APIs, PostgreSQL, Docker, Git / GitHub, Linux, Jupyter, Kubernetes

Additional Employment: Lifeguarding and Swim Instruction @ Oregon State Aquatics Department, 4yrs